

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented structural analysis workflows for reinforced concrete and steel elements using **GSA**, reducing analysis iteration time by approximately **20%**.
- Prepared detailed mark-ups for structural technicians, guiding the production of construction drawings in **Revit** for commercial building projects.
- Conducted on-site structural investigations and surveys, documenting findings and generating comprehensive **site reports** for client review.
- Represented the engineering team in project coordination meetings with clients and external consultants, clearly communicating technical designs and progress.
- Developed Python scripts to automate data extraction and reporting from engineering databases, streamlining the creation of **project status summaries**.
- Managed change control processes for system configuration updates, coordinating User Acceptance Testing (UAT) and providing post-go-live support for enterprise systems.
- Containerized Java Spring Boot applications using **Docker** and deployed them to **AWS EC2** instances, improving deployment consistency across development and production environments.

SKILLS

Structural Analysis Software – GSA, Tekla Structural Designer

CAD & BIM Software – Revit

Programming & Scripting – Python, Java

Cloud & DevOps – AWS (EC2, S3, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Engineering Concepts – Structural Design (Reinforced Concrete, Steel, Timber, Masonry), Site Investigations, Report Writing, Data Modelling

PROJECTS

Automated Structural Analysis Reporting (HCL Technologies)

- Developed a Python-based tool to parse output files from structural analysis software, extracting key load cases and deflection data for **commercial building projects**.
- Integrated the tool with **PostgreSQL** to store historical analysis results, enabling trend analysis and comparison across different design iterations.
- Generated automated summary reports in PDF format, highlighting critical structural parameters and potential areas for optimization, reducing manual reporting effort by **40%**.

Revit Drawing Generation Enhancement (HCL Technologies)

- Created Python scripts to automate the generation of repetitive structural elements in **Revit**, such as standard beam connections and column grids.
- Collaborated with structural technicians to refine the script outputs, ensuring adherence to **engineering drawing standards** and project specifications.
- Reduced the time spent on creating standard drawing elements by an estimated **25%**, allowing technicians to focus on more complex design aspects.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)